



NATURAL LANGUAGE PROCESSING · PROJECT PRESENTATION

Cyberpunk 2077:

What 92,000 Steam Reviews Reveal

15 NLP methods from tokenization to Retrieval-Augmented Generation applied to real player reviews

15 methods · 92,190 reviews

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One Question, 92,190 Reviews



Research question

Now that Cyberpunk 2077 is one of Steam's highly-rated games what do players praise, what do they still complain about, and can we answer questions about it directly from the reviews?



92,190

English reviews
via Steam API



94.8%

thumbs-up rate



24 mo

Jul 2024 – Jun 2026
post-recovery era



5

fields per review

Takeaway A post-recovery snapshot with free sentiment labels (thumbs up / down) ideal for supervised NLP.

Web Crawling: Static CSV + Live API



HOW IT WORKS

- Merge + deduplicate on every notebook run
- Dataset is always current (last fetch: Jul 2026)

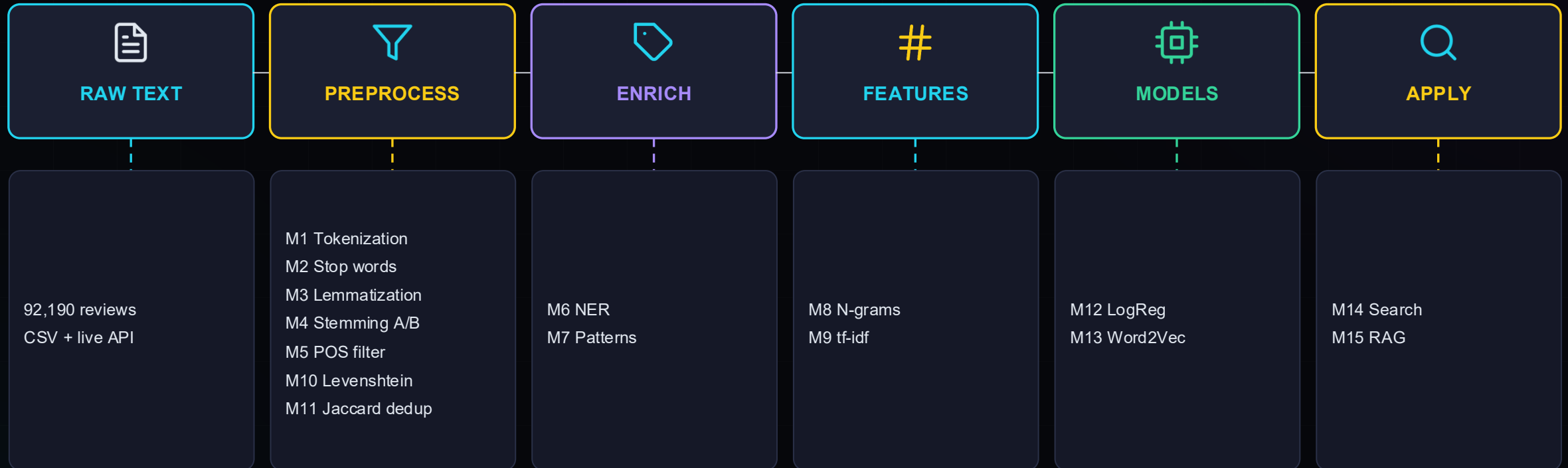


LIVE SAMPLE

*"game good. sometimes crashes and has bugs
but its worth it"*

Takeaway API-based collection is the modern form of web crawling the corpus refreshes itself.

One Pipeline, All 15 Methods



Takeaway Every method feeds the next: Levenshtein fixes typos before tf-idf counts them, Word2Vec expands queries for search, and search becomes the “R” in RAG.

Tokenize → Drop Stop Words → Lemmatize



M1 · Tokenization

text → tokens ("PS5", "10/10", emoji handled)



M2 · Stop-word removal

drop noise but keep "not" (negation!)



M3 · Lemmatization

crashes → crash · are → be (real words only)



One `nlp.pipe()` pass = tokens + lemmas + POS + entities · 92k reviews in minutes

ONE REVIEW THROUGH THE PIPELINE

"The graphics are amazingly beautiful but the game constantly crashes on my PS5."



graphic amazingly beautiful game constantly crash
ps

14 → 7

tokens kept

1

tf-idf column per lemma

"not"

always preserved

Takeaway 50% smaller text, meaning intact and negation survives by design.

Four Methods That Guard Data Quality



M4 · Stemming vs Lemma

Δ 0.06 pp

94.33% vs 94.27% we keep lemmas (readable)



M5 · POS filter

5 POS

NOUN · PROPN · VERB · ADJ · ADV (+ “not”)



M10 · Levenshtein fix

832

typos merged BEFORE tf-idf (beatiful→beautiful)



M11 · Jaccard dedup

−10,308

duplicates removed → 62,368 clean reviews

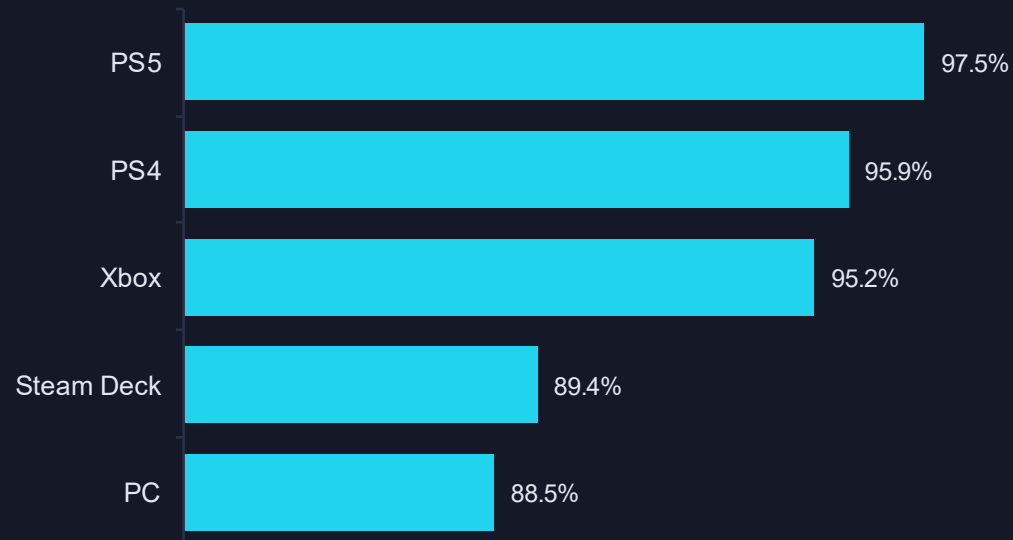
Takeaway Every cleaning decision was tested, not assumed A/B accuracy, edit distance, trigram similarity.

NER & Pattern Matching Create New Columns



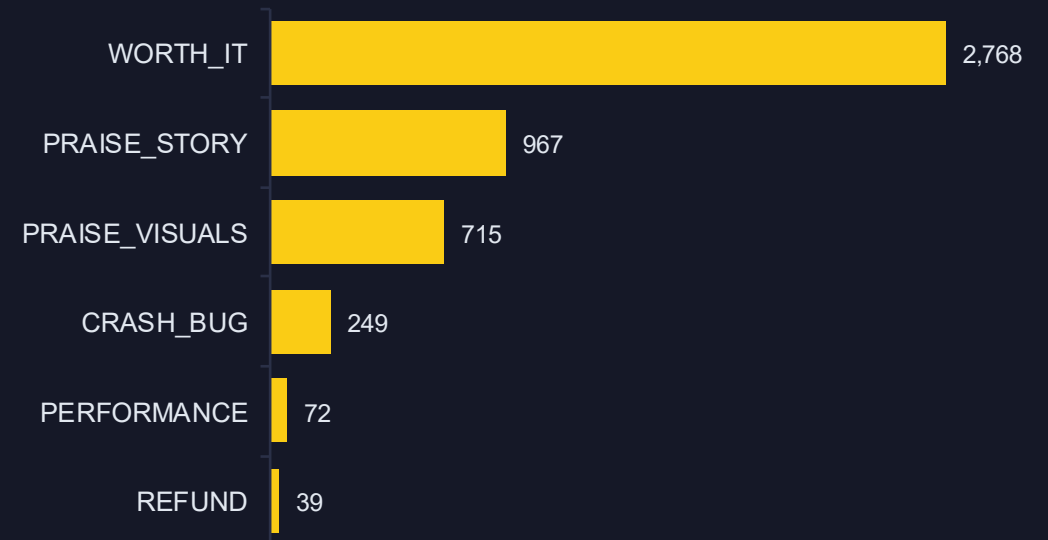
M6 · NER → platform column

% positive by platform



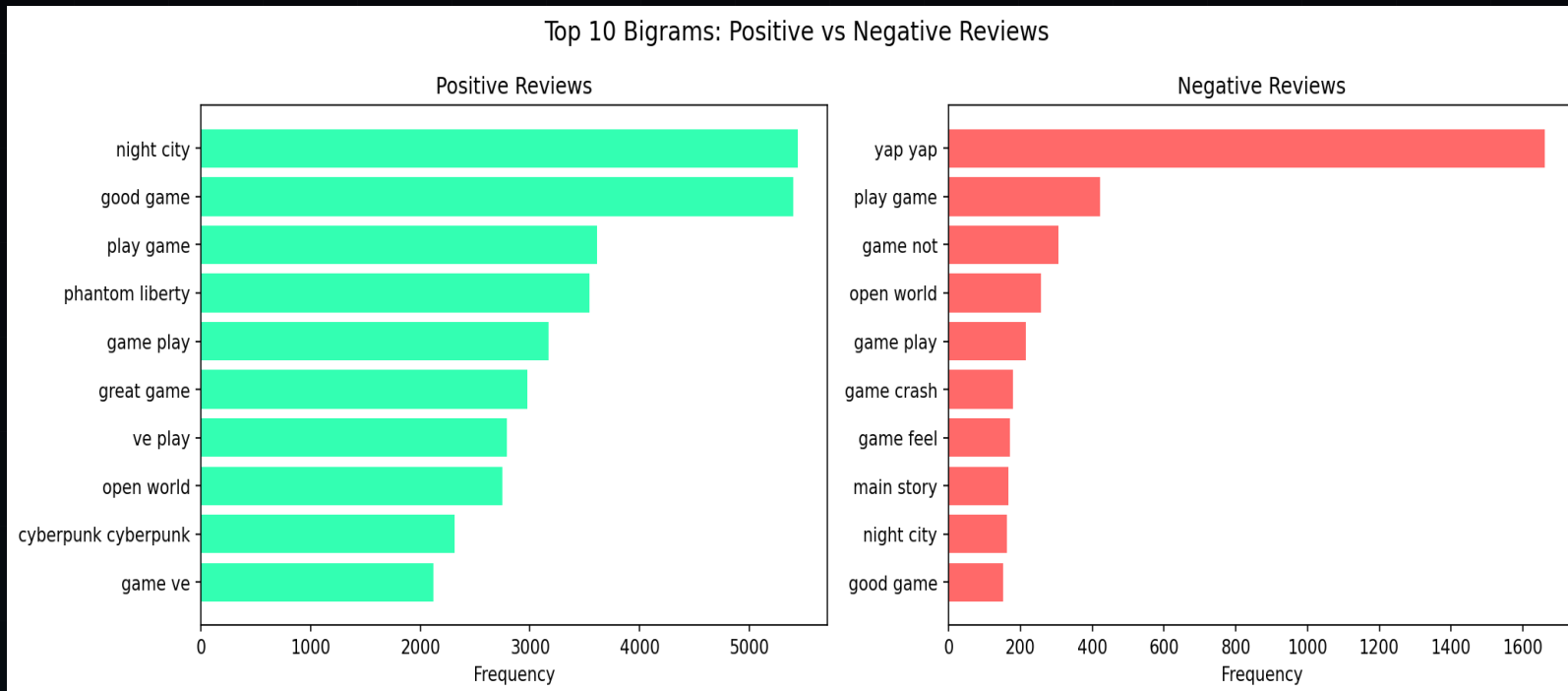
M7 · Matcher → category column

reviews per pattern tag (4,819 tagged)



Takeaway PS5 players are the happiest (97.5%), PC the most critical (88.5%) and praise tags outnumber complaint tags ~10 : 1.

N-grams: The Phrases Behind the Votes



KEY BIGRAMS

night city 5,444×

phantom liberty 3,546×

game crash 180×

yap yap (meme) 1,662×

Takeaway With ngram_range=(1,2) bigrams become tf-idf features the model can tell “not worth” from “worth”.

tf-idf: Rare = Informative



62,368 docs

cleaned reviews



$TF \times \log(N/df)$

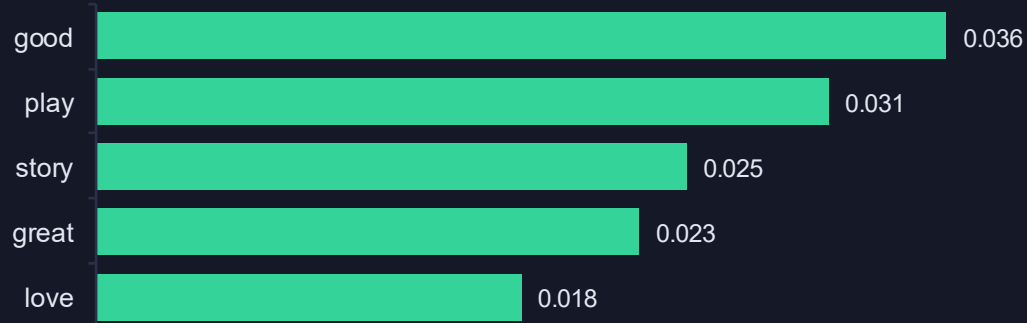
the lecture formula



$62,368 \times 8,000$

sparse matrix · 99.7% zero

Top positive-class terms (mean tf-idf)



Top negative-class terms (mean tf-idf)



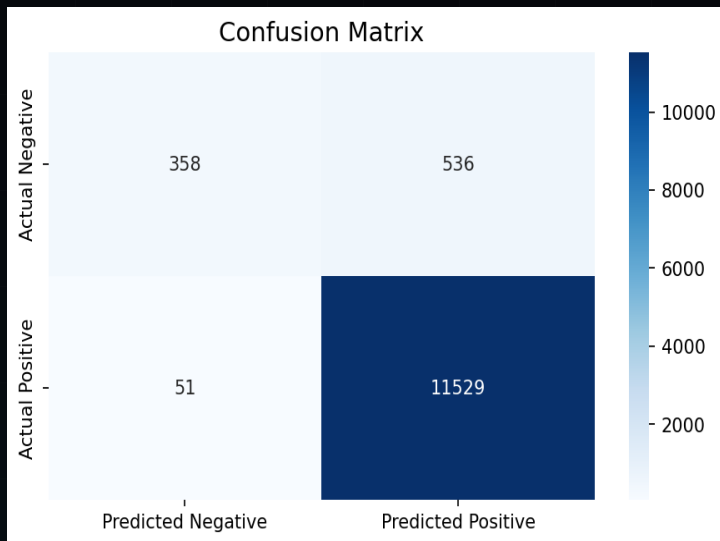
Takeaway “not” is the 2nd-strongest negative term protecting it from stop-word removal paid off.

Sentiment Classification with Logistic Regression



95.3%

test accuracy · 12,474 reviews



WHAT THE MODEL LEARNED LOGISTIC REGRESSION COEFFICIENTS

Positive drivers ▲



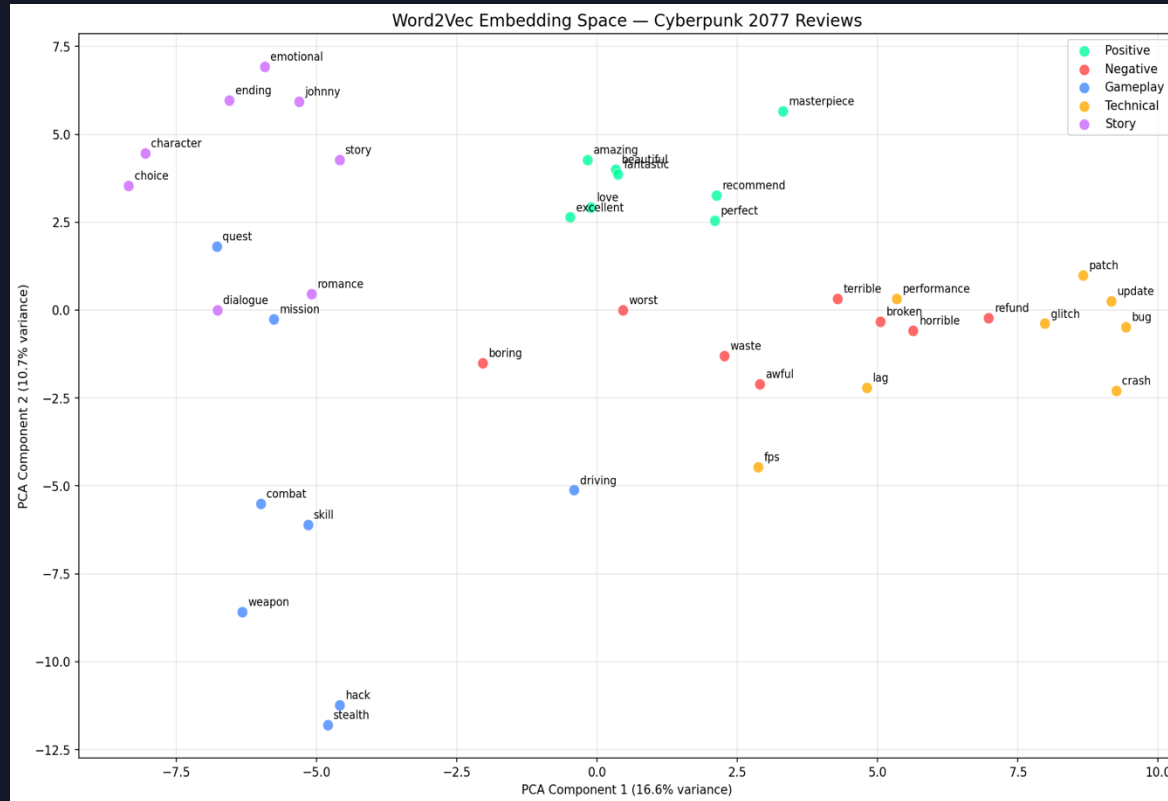
Negative drivers ▼



“choom” = Cyberpunk slang for friend the model learned the game's own language.

Takeaway Fully interpretable but 95/5 class imbalance caps negative recall at 0.40 (a data property, not a model failure).

Word2Vec: Semantic Clusters from Context Alone



PCA check: positive & negative words separate in space
6,115 words · 100-D · CBOW

Technical issues

bug

glitch .85

crash

freeze .69

lag

Praise

beautiful

gorgeous .80

stunning .67

breathtaking

Narrative

story

storyline .83

plot .67

narrative

VECTOR ARITHMETIC

$\text{bug} + \text{fix} - \text{crash} \approx \text{patch}$
 $\text{graphic} + \text{good} - \text{bad} \approx \text{amazing}$
 $\text{story} + \text{excellent} - \text{terrible} \approx \text{storyline}$

Takeaway Trained on our own corpus it even clusters typos (“reccomend”) with their correct forms.

A Search Engine Over 62,000 Reviews



EXAMPLE QUERY EXPANSION IN ACTION

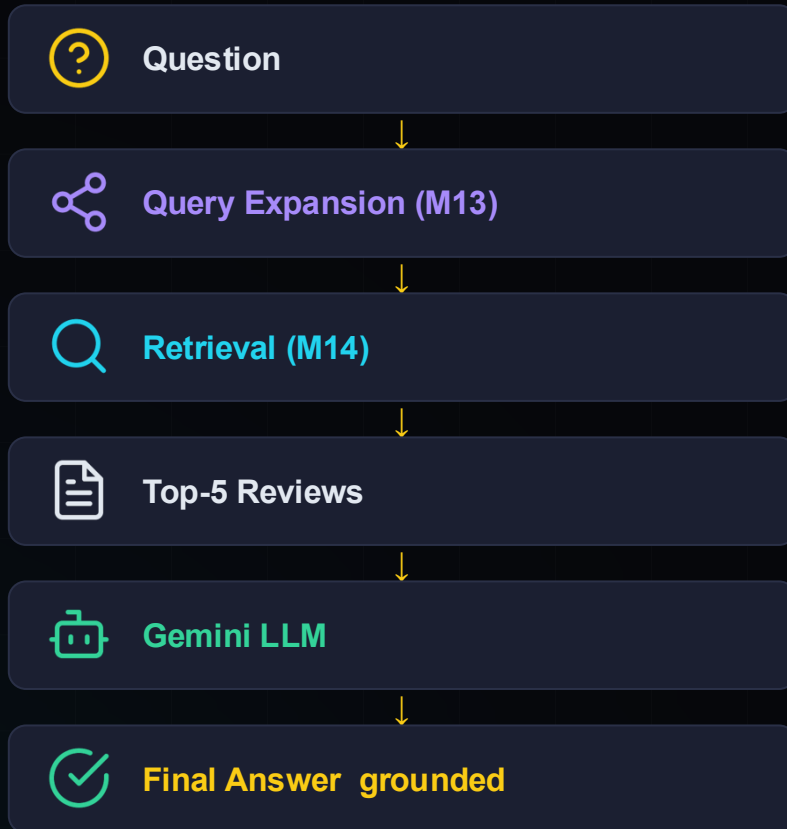
Query "How is the story and characters?"

Expanded + storyline · plot · plotline

Top hit "Really enjoying the plotline so far!" (cos 0.60)

Takeaway Embeddings bridge the vocabulary gap ("plot" $\approx$ "story") this engine becomes the "R" in RAG.

RAG: Answers Grounded in Real Reviews



WORKED EXAMPLE

Q “Does the game still have bugs and crashes?”

Retrieved evidence (top-5):

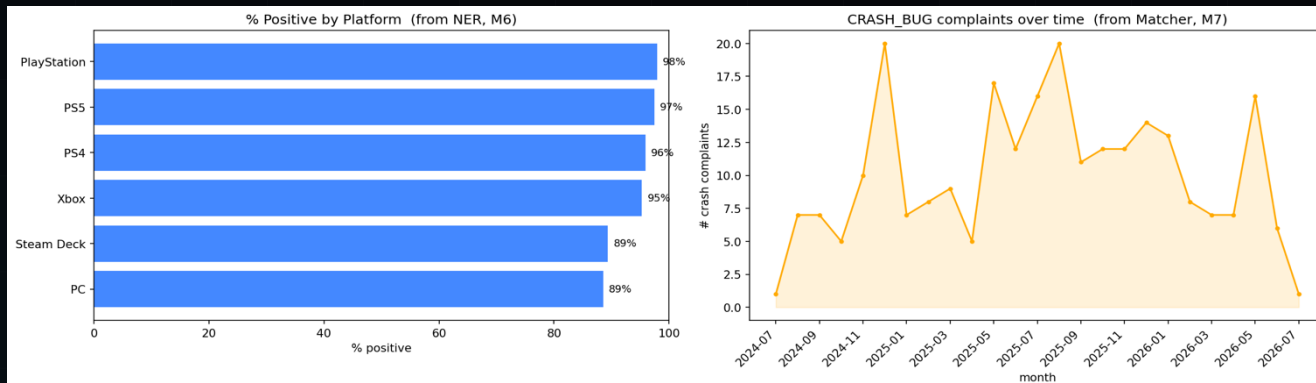
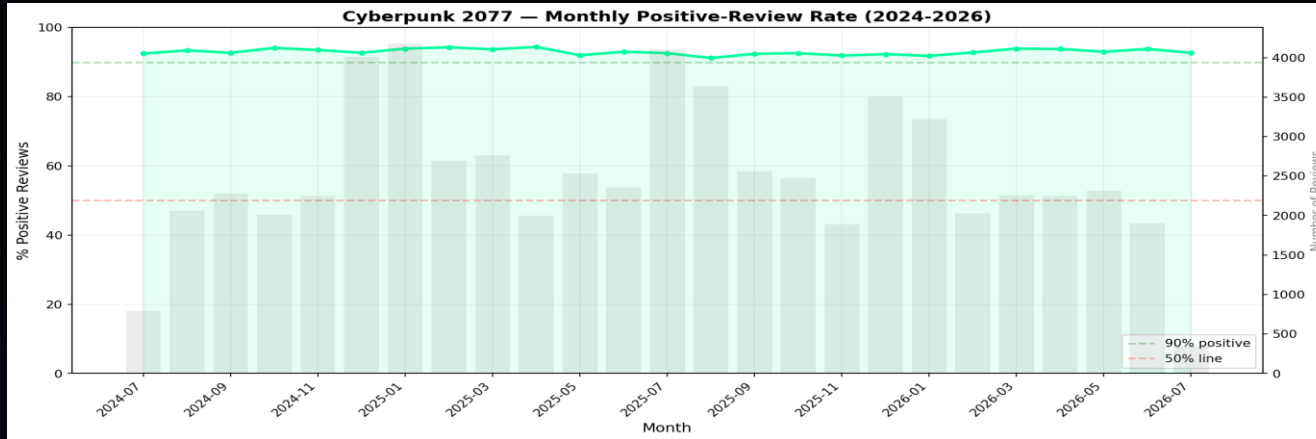
“Bugs: 10/10, Game: 8/10” · “Must play experience...” · 3 more (incl. non-English, still ranked correctly)

Prompt rule: “Answer *ONLY* from these reviews” → no hallucination

A (grounded) “Yes several players still report bugs, and one warns they can hurt the experience. But even those reviewers rate the game positively and recommend it overall.”

Takeaway Every claim in the answer is traceable to a retrieved review retrieval + generation, no hallucination.

Findings: Stable Praise, Persistent Complaints



92.7%

positive in Jul 2026 vs 92.5% in Jul 2024. Two years, no decline.



97.5 vs 88.5

% positive: PS5 vs PC the platform gap (NER column).



249

CRASH_BUG mentions rare but never zero (Matcher column).

Takeaway Sentiment is high AND stable the recovery holds; crashes remain the last stubborn weakness.

15 Methods, One Pipeline, One Answer



RESEARCH QUESTION

What do players praise & still complain about and can we answer questions directly from the reviews?



MAIN FINDINGS

- 92–93% positive, stable for 2 years
- Praise: Night City · story · Phantom Liberty
- Complaints: crashes · rare but persistent
- RAG answers questions grounded in reviews



LIMITATIONS

- English-only corpus



FUTURE WORK

- Class weights / SMOTE for imbalance
- Multilingual models (incl. German reviews)
- Larger retrieval context · transformer embeddings